



AI-powered GPS tool called ORION (on road integrated optimisation and navigation) to create most efficient routes for fleets (Credit: <https://i.pinimg.com>)

A number of postal services such as US Federal Post Agency, Russian Post, and Australian State Post are slowly showing interest in blockchain technology.

The IoT, AI and robotics are the future of logistics

Nowadays, many different kinds of delivery companies transport different kinds of parcels and offer their own services, which causes a lot of wastage of resources. In addition, volume of parcels in all cities that need to be delivered has grown dramatically. To deal with these problems, a uniform parcel delivery system in a smart city that can offer service to all kinds of customers in the city, including manufacturers, departmental stores, restaurants, individuals and so forth, has been designed. This system uses the IoT, RFID technology, wireless communication and cloud computing. With this system, the whole package delivery process, including classification of packages, vehicle scheduling, path planning, transporting and monitoring, can be managed automatically. As a result, use of both material and manpower resources can be reduced.

Artificial intelligence (AI) has the power to increase efficiency of the

supply chain in all areas drastically. Rolls Royce recently partnered with Google to create autonomous ships. This has resulted in replacement of jobs of twenty-plus ship crew members.

In supply chain deliveries, every minute and kilometre matters. UPS, a parcel delivery company, uses an AI-powered GPS tool called ORION (on road integrated optimisation and navigation) to create the most efficient routes for its fleet. Customers, drivers and vehicles submit data to the machine, which then uses algorithms to create the most optimal routes. Instead of being stuck in traffic, ORION helps drivers make their deliveries on time and in the most efficient manner.

Optimising delivery routes has had a great impact on UPS business, from saving time and money to reducing emissions and wear-and-tear of its trucks. With ORION, UPS estimates to have reduced its delivery kilometres by hundred million. Thus, using AI, UPS is saving fifty million dollars per year.

Smart logistics refers to the realisation of system perception, comprehensive analysis and processing in various kinds of logistics, including storage and distribution. Use of the IoT technology can promote intelligent levels of the whole logistics industry.

In south England, a large group of robots occupy a three-storied grocery fulfilment centre, managing 50,000 products in three temporary regimes, each fulfilling customer orders, communicating over wireless networks.

SaaS parcel shipping solutions used to be only for small shipping, but today, the IoT connects shipping apps to local data sources, scales, thermal label printers and other material-handling devices.

The IoT is supporting the value chain in parcel delivery to create an Internet of parcels. Digitalisation and the IoT are creating new framework conditions. The domestic market of US Postal Service, China Post and Japan Post is big enough. The IoT can provide a sustainable solution for these to support the growing demand of e-commerce for seamless x-border delivery. Today, there is growing demand for x-border e-commerce, B2C delivery.

It is crucial that parcels arrive on time at the destination to satisfy customers. Parcel delivery systems involve many steps, including scanning parcel codes, loading parcels onto vehicles and sending them to the customers' destination. Today, GPS, Waze, Android tools and other cloud-based systems are intelligently managing delivery processes and delivering parcels on time. Several new technologies are being used as e-commerce and online shopping is exploding.

According to a tech company, delivery robots are being designed to deliver packages, groceries and food to consumers within two-to-three-kilometre radius in under thirty minutes. These drive autonomously while being monitored by human operators in control centres. The robots have already been on the road in European and American cities and have covered more than 8050km (5000 miles), meeting more than 400,000 people without a single accident.

Building a robust courier industry needs long-term planning. Like any other industry that is at a nascent stage, the electronics-based parcel industry in India will also undergo its share of teething problems. This is where a healthy interface between the government and courier companies will ensure that businesses scale rapidly in the country. **EFY**